# Slovak Technical University in Bratislava

Faculty of Informatics and Information Technologies


# Developing Neuron Network Architecture for MAGIC Gamma Telescope

Introduction to Neural Networks


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**Time of the practice**: Monday 15:00 - 16:50

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# Introduction

## Project Overview

This project implements a machine learning pipeline to classify high-energy gamma particles from background hadron events using data from the MAGIC (Major Atmospheric Gamma Imaging Cherenkov) telescope. The classification task is important for distinguishing signal (gamma rays) from background noise (cosmic rays) in atmospheric Cherenkov telescope observations.

## Dataset Information

Source: UCI Machine Learning Repository

(MAGIC Gamma Telescope - UCI Machine Learning Repository)

Instances: 19,020 observations

Features: 10 continuous features derived from shower image parameters

Target: Binary classification (gamma and hadron)

Gamma (signal): 12,332 instances

Hadron (background): 6,688 instances

## Work Distribution

Filip Brutovský (xbrutovskyf@stuba.sk): 50%

Anastasiia Rerikh (xrerikh@stuba.sk): 50%

## Libraries used:

pandas

numpy

matplotlib

seaborn

scipy

torch

sklearn

ucimlrepo

wanb

itertools

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# Data Preprocessing

## Initial Data Exploration

### Feature Descriptions:

1. ***fLength***: Major axis of the ellipse - longest spread of light in the shower image. Indicates shower elongation. [mm]
2. ***fWidth***: Minor axis of the ellipse - width perpendicular to the major axis. [mm]
3. ***fSize:*** Logarithmic sum of all pixel intensities (brightness of event). Related to total energy.
4. ***fConc:*** Ratio of the two brightest pixels to total intensity - compactness measure.
5. ***fConc1:*** Ratio of brightest pixel to total intensity.
6. ***fAsym:*** Asymmetry along major axis - offset of light centroid. Indicates directionality. [mm]
7. ***fM3Long:*** Third root of third moment along major axis (shape moment). Describes tail shape. [mm]
8. ***fM3Trans:*** Third root of third moment along minor axis. [mm]
9. ***fAlpha:*** Angle of major axis vs. camera centre line (orientation). Used for source direction. [deg]
10. ***fDist:*** Distance from origin (camera centre) to ellipse centre — indicates offset of the shower. [mm]

### Anomalies Examples:

- Zero or negative geometry: fWidth = 0, fLength <=0, fSize <=0.
- Impossible angles: 0 < fAlpha, fAlpha > 180.
- Very large or small distances: fDist > 100, fDist < 0.

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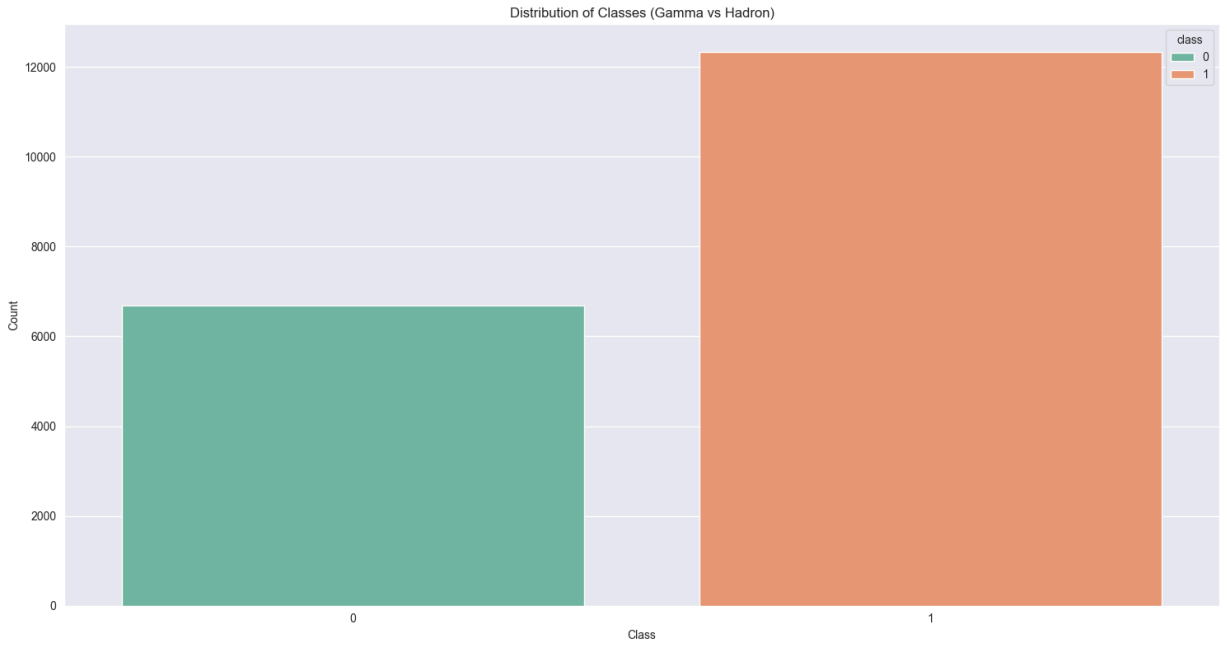
## Potential issues:

- 115 duplicate records identified;
- Class imbalance: ~65% gamma, ~35% hadron events;
- Missing values - 0;
- Anomaly values - 0;
- Outliers - 7011.




## Data Cleaning - Outlier Detection and Handling

Method: Interquartile Range (IQR) method

Outliers defined as values outside [Q1 - 1.5×IQR, Q3 + 1.5×IQR]

Outliers replaced with NaN, then imputed with the median.

Applied to all numerical features.

Missingness Handling:

Threshold: 30% missing values

Drop columns with >30% missing

Impute columns with 0.1-30% missing using the median

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## Normalisation and Scaling

We transform skewed distributions to be closer to the Gaussian. Normalisation and scaling have been applied to 7 features with significant skewness: *fLength, fWidth, fSize, fAlpha, fAsym, fM3Long* and *fM3Trans*.

On the other hand, features such as *fConc*, *fConc1* and *fDist* remain reserved as they were initially well-distributed.

## Target Variable Encoding

**1 - gamma;**

**0 - hadron.**

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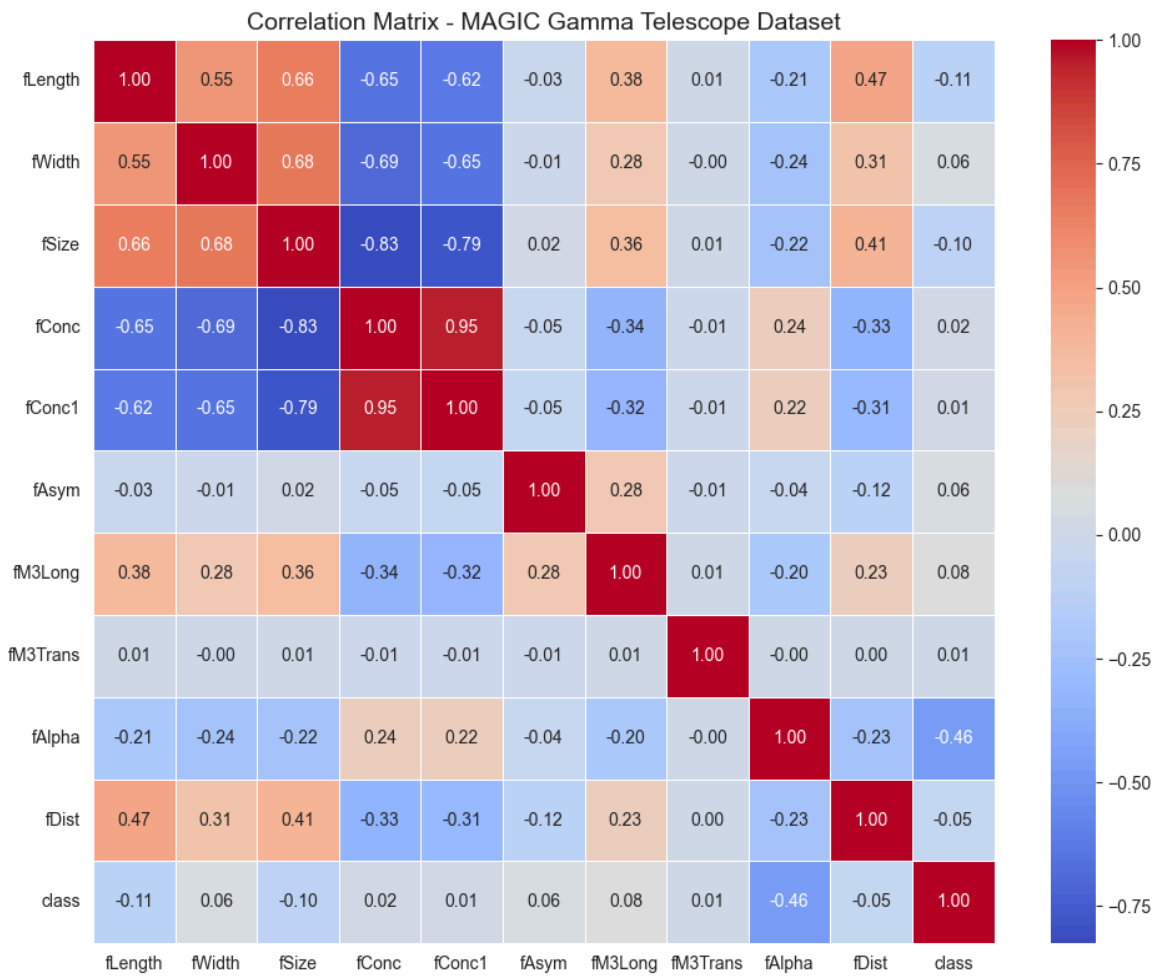
# Feature Selection

The goal of this selection is to identify which features in the dataset are the most informative for predicting the target variable. We applied a selection of techniques to do so, including: filtering method (Correlation analysis) and wrapped methods (Forward and Backwards Elimination).

## Correlation matrix



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# Forward Selection

## Pseudocode:

Start with no features. ($R^2 = 0.20$)

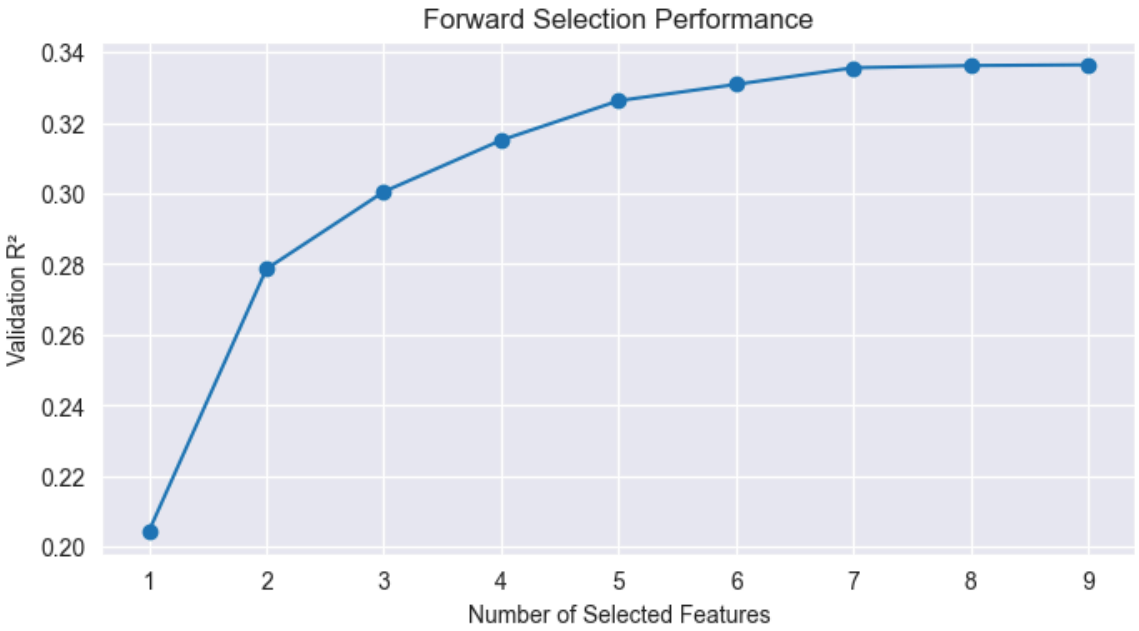
Train model → measure score.

Try adding each remaining feature one by one.

Add the feature that gives the best improvement.

Repeat until no feature improves the score.

## Results:




Selected Features (9): *fAlpha, fSize, fWidth, fLength, fConc, fM3Long, fDist, fConc1, fAsym*

Final Validation $R^2$: 0.3364

Excluded: *fM3Trans*

## Key Observations:

Most informative feature: *fAlpha* (angle of major axis)

Gradual performance improvement with each added feature

Minimal difference between 9 and 7 features

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# Backward Elimination

## Pseudocode:

Start with all features. (10 features; $R^2 = 0.3363$.)

Train model -> compute baseline score ($R^2$).

For each feature:

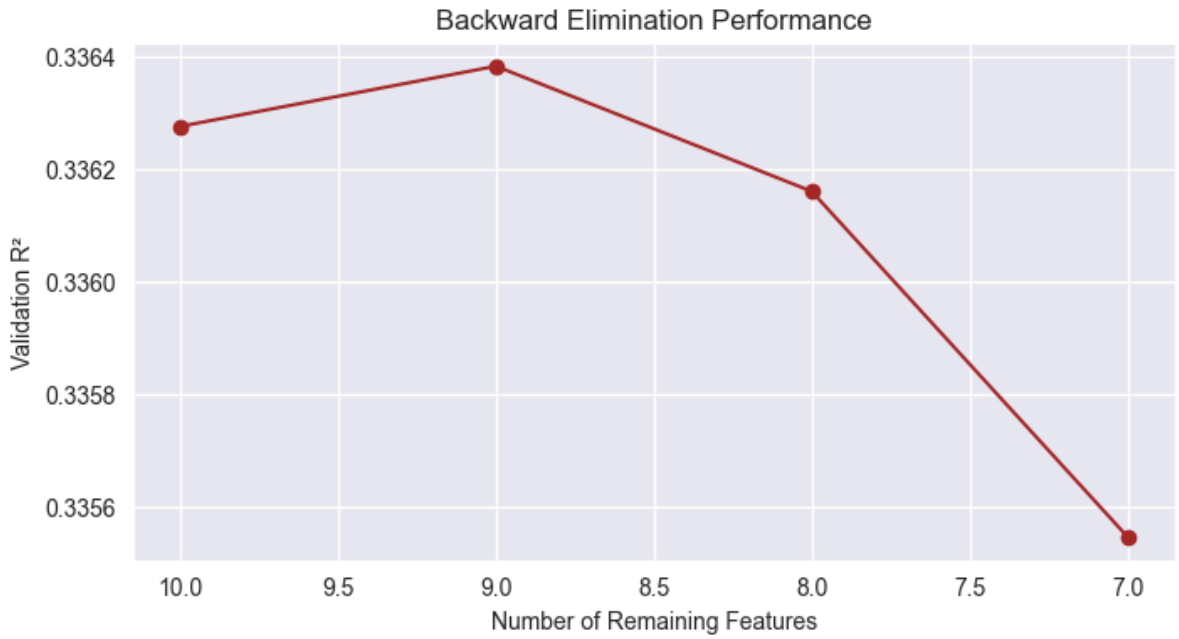
Temporarily remove it -> train the model again.

Record validation score.

Remove the feature whose removal causes the smallest drop (or largest gain) in score.

Repeat until the score drops or a stopping condition is met.

## Results:




Removal sequence:

Removed *fM3Trans* → $R^2 = 0.3364$ (slight improvement)

Removed *fAsym* → $R^2 = 0.3362$ (slight decrease)

Removed *fConc1* → $R^2 = 0.3355$ (decrease stops process)

Final selected features (7):

*fLength, fWidth, fSize, fConc, fM3Long, fAlpha, fDist*

Final $R^2$: 0.3364

## Feature Selection Conclusion

The two chosen methods have identified *fAlpha* as being the strongest predictor. The optimal feature set contains 9 features, excluding *fM3Trans*.

# Neural Network Overview

## Network Structures:

1. **Magical Net 1:**

   *Structure:*

   a. Input Layer (9 features)
   b. Hidden Layer 1: Linear(9 - 64) -> Tanh
   c. Hidden Layer 2: Linear(64 - 64) -> ReLU
   d. Output Layer: Linear(64 -> 1) -> Sigmoid (via BCEWithLogitsLoss)

   *Activation function:* ReLU, Tanh.

2. **Magical Net 2:**

   *Structure:*

   a. Input Layer (9 features)
   b. Hidden Layer 1: Linear(9 - 64) -> BatchNorm1d -> ReLU -> Dropout
   c. Hidden Layer 2: Linear(64 - 64) -> BatchNorm1d -> ReLU -> Dropout
   d. Output Layer: Linear(64 -> 1) -> Sigmoid (via BCEWithLogitsLoss)

   *Activation function:* ReLU.
   *Normalisation:* Batch Normalisation for training stability
   *Regularisation:* Dropout for preventing overfitting

3. **Magical Net 3:**

*Structure:*

a. Input Layer (9 features)
b. Hidden Layer 1: Linear(9 - 64) -> BatchNorm1d -> Leaky ReLU -> Dropout
c. Hidden Layer 2: Linear(64 - 64) -> BatchNorm1d -> Leaky ReLU -> Dropout
d. Hidden Layer 3: Linear(64 - 64) -> BatchNorm1d ->Leaky ReLU -> Dropout
e. Output Layer: Linear(64 -> 1) -> Sigmoid (via BCEWithLogitsLoss)

*Activation function:* Leaky ReLU.

*Normalisation:* Batch Normalisation for training stability

*Regularisation:* Dropout for preventing overfitting

4. **Magical Net 4:**

*Structure:*

a. Input Layer (9 features)
b. Hidden Layer 1: Linear(9 - 128) -> BatchNorm1d -> Leaky ReLU -> Dropout
c. Hidden Layer 2: Linear(128 - 64) -> BatchNorm1d -> Leaky ReLU -> Dropout
d. Output Layer: Linear(64 -> 1) -> Sigmoid (via BCEWithLogitsLoss)

*Activation function:* Leaky ReLU.

*Normalisation:* Batch Normalisation for training stability

*Regularisation:* Dropout for preventing overfitting

# Training Configuration

## Data split:

Train size/ Test size: 80% (15,216 samples) / 20% (3,804 samples)

Stratification: Preserves class distribution in both sets

Random state: 14(reproducibility)

## Hyperparameters:

Batch Size: 256, 512

Learning Rate: 1e-3

Optimiser: AdamW with weight decay

Loss Function: BCEWithLogitsLoss with pos_weight=1.8

Epochs: 500 (with early stopping)

- Experimented with 100, 500, 1000 - after experiments, we got to the conclusion that the best value/cost metric had 500 epochs.

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# Experiments with the network

## Network depth

- Two hidden-layer structure - initial width 64 then adjusted to 128 for better results.
- Three hidden-layer structure - improved metrics but it was prone to overfitting.

## Activation functions

- Tanh - saturated our data and was not a good fit for our NN.
- ReLU - simple and effective, but our data had some negative values, and by using ReLU, we lost information about them.
- Leaky ReLu - Modified version of ReLu, it fits our problem like a glove, it prevents the loss of information problem we had by using simple ReLu.

## Improvement Techniques

### Dropout

- 0.2 dropout rate (20%) Had a positive effect on the NN's performance and prevented overfitting.
- 0.3 dropout rate (30%) Had negative effect, too much information was discarted

### Normalisation layers

- After we introduced the Normalisation layers to our NN. It stabilised the learning curve, and we could try greater learning rates.

### Loss-function parameter tuning

binary cross-entropy (pos_weight)

- 2.0 - Had a moderate improvement level in contrast to 1
- 3.0 - Was overkill, it hindered the NN's accuracy and was punishing the error too much
- 1.5 - Improved accuracy and f1-score considerably
- 1.8 - After more tuning (Grid search), we settled for this value, and it improved a little in comparison to the 1.5's results.

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## Hyperparameter search

Grid search

Once the new feature was added, we reran the grid search to tune its associated parameter. This allowed us to determine the optimal value for the new feature and evaluate its impact on overall model performance. Its epoch rate was scaled down in order to find values with a great start and after that we used a normal training method with more epochs in order to find the best possible combination.

## Learning Rate Schedule

A learning rate scheduling strategy was introduced to improve training stability and convergence. However, the results showed no significant improvement in performance, indicating that the learning rate schedule was ineffective for this particular setup.

## Early stopping

Early stopping was introduced to prevent overfitting and to catch more promising instances of NN by monitoring validation performance during training. It proved highly effective in stabilising results. Also, it helped with reducing the number of false positive results by focusing on the F1-score.

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# Results and evaluation

## Best Network Performance

Name of the network: MagicalNet 4

Training completed: 500 epochs

Best epoch: 500

## Final Validation Metrics (for the best)

Accuracy: 87.5%

Precision: 85.61%

Recall: 96.43%

F1-Score: 91.35%

## Confusion Matrix

| | Predicted Negative | Predicted Positive |
|---|---|---|
| Actual Negative | 892 | 397 |
| Actual Positive | 84 | 2431 |

Interpretation:

**True Negatives (TN):** x hadrons correctly classified

**False Positives (FP):** x hadrons misclassified as gamma

**False Negatives (FN):** x gammas misclassified as hadron

**True Positives (TP):** x gammas correctly classified

However, it is important to notice that **false positives** (hardon -> gamma) are less severe as they only waste the telescope time, whereas **false negatives** (gamma -> hardon) may cause the overlook of scientifically important events.

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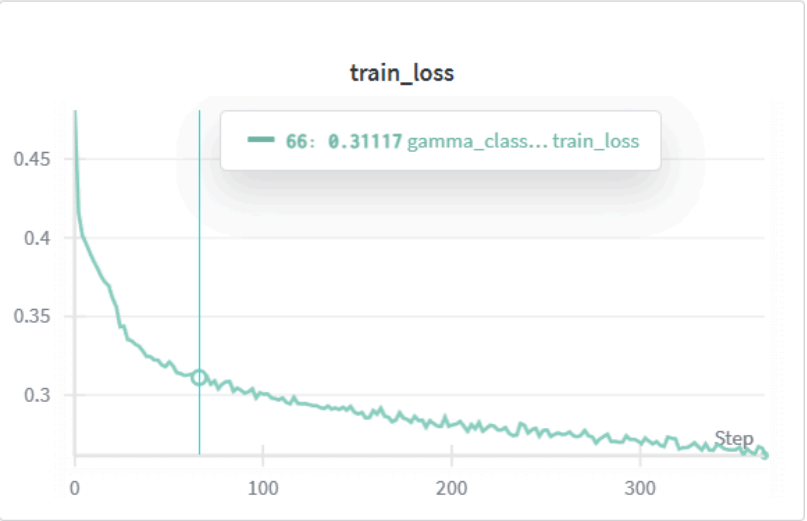
# Training History Analysis

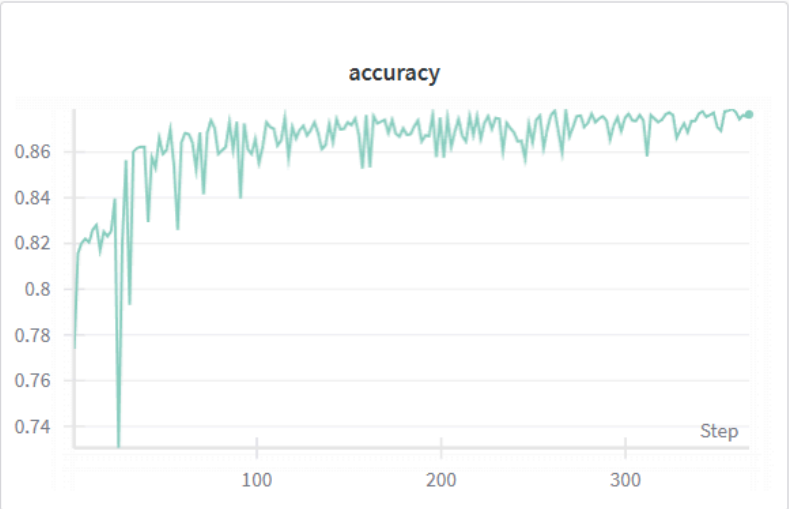
**Loss Trajectory:**

- Epoch 1: 1 -> Epoch 500:
- Smooth, monotonic decrease
- No signs of overfitting



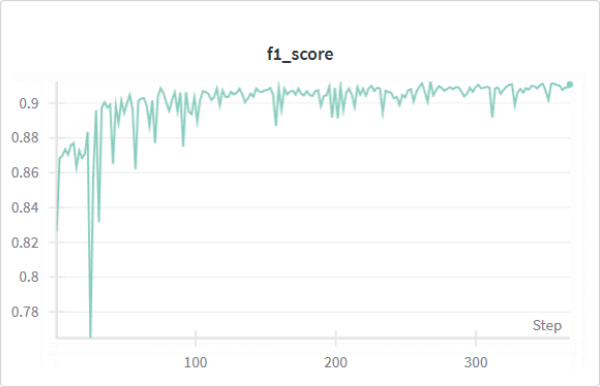

**Accuracy Progression:**

- Epoch 1: -> Epoch 500:
- Peak performance at final epoch



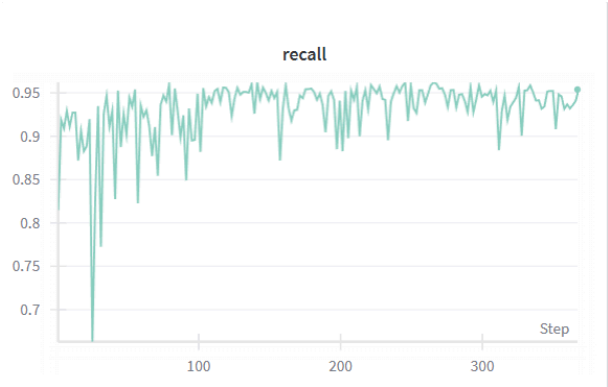
**F1 Progression:**

- Epoch 1: -> Epoch 500:
- Peak performance at final epoch



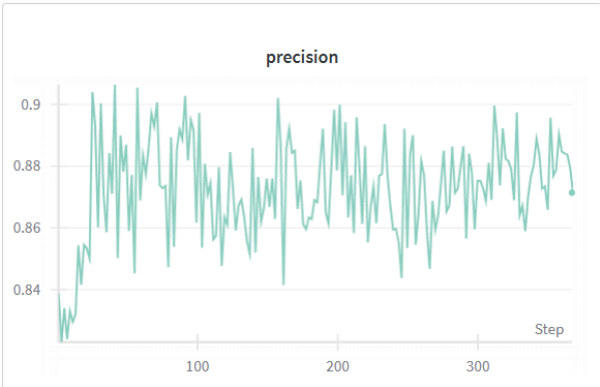

**Recall Progression:**

- Epoch 1: -> Epoch 500:
- Peak performance at final epoch



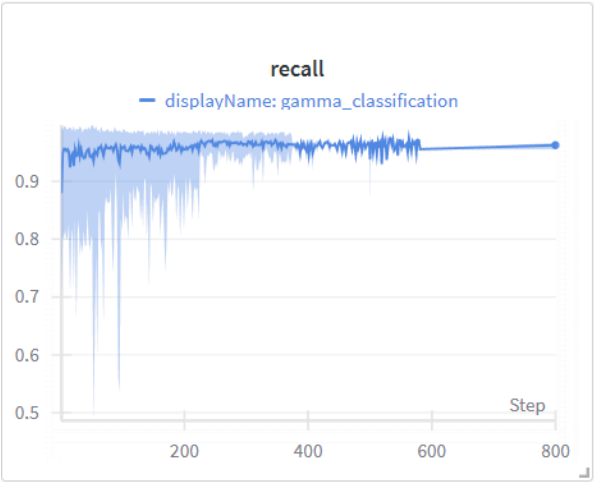

**Precision Progression:**

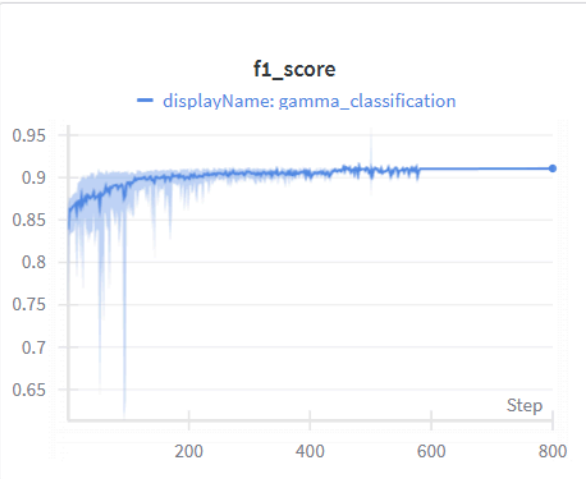
- Epoch 1: -> Epoch 500:
- Peak performance at final epoch

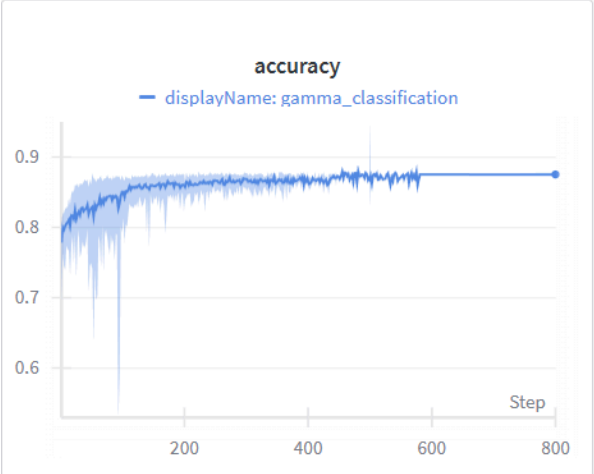


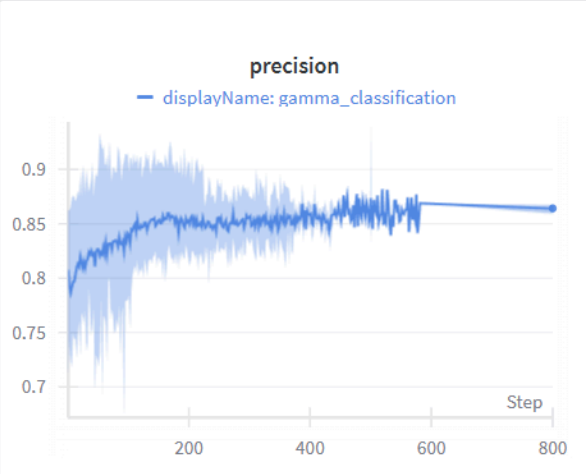
# Comparative Analysis

## Graphs




## Experimental conclusion

Throughout model development, several improvements and techniques were tested to enhance performance. Weighted loss (pos_weight) was adjusted to account for class imbalance, improving the balance between precision and recall. Normalisation layers stabilised training and accelerated convergence, while dropout, when applied, improved performance; without it NN had problems with generalisation. Early stopping was effective in catching unstable learning progress. Among activation functions, Leaky ReLU outperformed Tanh and ReLU by preventing dead neurons and maintaining stable gradient

flow. Learning rate scheduling, however, did not provide additional benefits in this setup. Taken together, these experiments demonstrate that careful handling of class imbalance, normalisation, and activation selection had the most significant positive impact on model performance.

# Overall Conclusion

This project successfully developed a neural network classifier for distinguishing gamma-ray events from background hadron noise in MAGIC telescope data. The final model, MagicalNet 4, achieved 87.5% accuracy with a recall of 96.43% and an F1-score of 91.35%, demonstrating strong performance.

Through systematic experimentation, several key findings emerged: Leaky ReLU activation functions outperformed Tanh and standard ReLU by preventing information loss; batch normalisation and 20% dropout effectively stabilised training and prevented overfitting; weighted loss functions (pos_weight=1.8) successfully addressed class imbalance; and feature selection identified fAlpha as the strongest predictor among 9 informative features.

The model's high recall (96.43%) is particularly valuable for gamma-ray astronomy, ensuring that scientifically important events are rarely missed. The precision of 85.61% results in some false positives, but these are acceptable as they only waste telescope time rather than cause missed discoveries.